Arjuna JEE 2027

Physics by Rajwant Singh Sir

Units and Measurements

DPP: 11

Time duration - 30 min

Q1 The most accurate reading of the length of $6.28\ cm$ long fibre is
(A) $6\ cm$
(B) $6.5\ cm$
(C) $5.99\ cm$
(D) $6.0\ cm$

Q2 The number of significant figures in 0.06900 is:
(A) 5 (B) 4
(C) 2 (D) 3

Q3 The number of significant figures in $0.0006032\ m^2$ is:
(A) 4 (B) 5
(C) 7 (D) 3

Q4 What is the number of significant figures in 0.310×10^3 ?
(A) 2 (B) 3
(C) 4 (D) 3

Q5 The sum of the numbers $436.32, 227.2$ and 0.301 in appropriate significant figures is:
(A) 663.821 (B) 664
(C) 663.8 (D) 663.82

Q6 Taking into account the significant figures, what is the value of $(9.99\ m - 0.0099\ m)$?
(A) $9.98\ m$
(B) $9.980\ m$
(C) $9.9\ m$
(D) $9.9801\ m$

Q7 The decimal equivalent of $1/20$ upto three significant figures is
(A) 0.0500
(B) 0.05000
(C) 0.0050
(D) 5.0×10

Q8 The numbers 2.745 and 2.735 on rounding off to 3 significant figures will give:
(A) 2.75 and 2.74
(B) 2.74 and 2.73
(C) 2.75 and 2.73
(D) 2.74 and 2.74

Q9 The length, breadth, and thickness of a block are given by $l = 12\ cm$, $b = 6\ cm$ and $t = 2.45\ cm$. The volume of the block according to the idea of significant figures should be:
(A) $1 \times 10\ cm^3$
(B) $2 \times 10^2\ cm^3$
(C) $1.764 \times 10\ cm^3$
(D) None of these

Q10 The mass and volume of a body are $4.237\ g$ and $2.5\ cm^3$, respectively. The density of the material of the body in correct significant figures is:
(A) $1.6048\ g\ cm^{-3}$
(B) $1.69\ g\ cm^{-3}$
(C) $1.7\ g\ cm^{-3}$
(D) $1.695\ g\ cm^{-3}$

Q11 Subtract 2.5×10^6 from 4.0×10^{-4} with due regard to significant figures
(A) 4.0×10^{-4}
(B) 6.0×10^4
(C) 4.0×10^{-3}
(D) 4.0×10^5

Q12 The mass and radius of the earth are $5.975 \times 10^{24}\ kg$ and $6.37 \times 10^6\ m$

(A) 5.53×10^3
(B) 5.52×10^3
(C) 5.54×10^3
(D) 5.5×10^3

respectively. Calculate the average earth's density to correct significant figures. Take pi = 3.142

(A) 5.53×10^{3}

(B) 5.52×10^{3}

(C) 5.54×10^{3}

(D) 5.5×10^{3}

Q13 The significant figures of the number 6.0023 are:

(A) 1 (B) 5

(C) 4 (D) 2

Q14 If $L\ 2.331\ cm = B\ 2.1\ cm$ = then $L + B$ =

(A) 4.423 cm

(B) 4.43 cm

(C) 4.4 cm

(D) 4 cm

Answer Key

Q1 (B)

Q2 (B)

Q3 (A)

Q4 (B)

Q5 (C)

Q6 (A)

Q7 (A)

Q8 (D)

Q9 (B)

Q10(C)

Q11(A)

Q12(B)

Q13(B)

Q14(C)